



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO Sree Malaimathi Agency

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

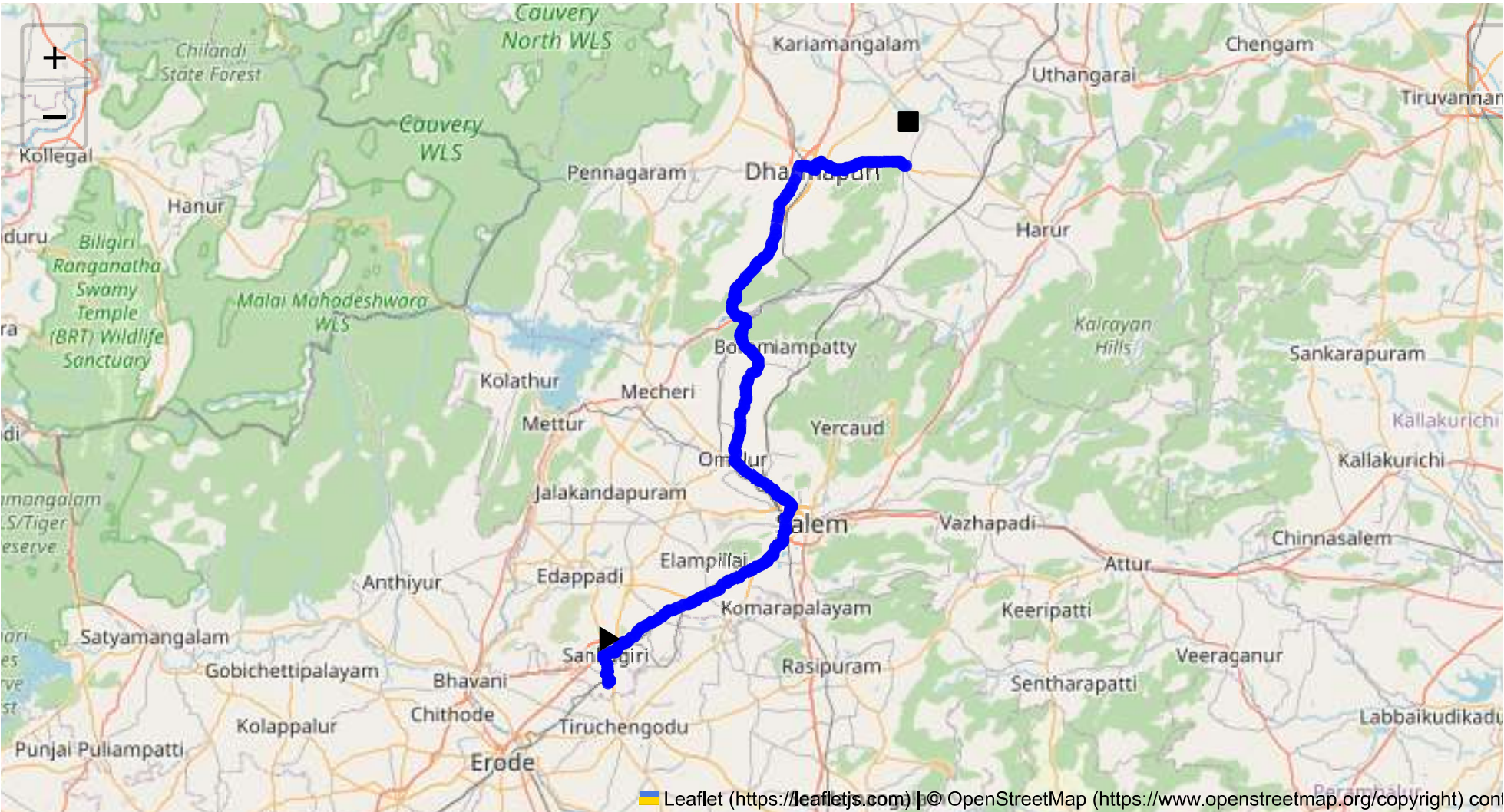
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 123.04 km**  
**Estimated Duration: 2.4 hours**  
**Adjusted Duration (Heavy Vehicle): 3.0 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.141649, 78.287325)**



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

- The route primarily follows NH79, SH86, and various local roads. It traverses rural and semi-urban areas, passing through towns like Sankari, Salem, and Dharmapuri.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards:

- The area typically experiences a tropical climate. The dry season lasts from January to April, which is ideal for transport. Monsoon season, from June to September, can result in heavy rainfall, leading to slippery roads, reduced visibility, and potential flooding in low-lying areas. Preparations should be made for these conditions, such as ensuring functional windshield wipers and adequate tire tread.

## 3. Analysis of Traffic Patterns:

- Major congestion can occur in Salem, a significant urban area along the route. Peak traffic hours are usually 8-10 AM and 6-8 PM during weekdays. Additional congestion can occur near toll plazas, especially during weekends and holidays.

## 4. Assessment of Road Quality and Infrastructure:

- NH79 and SH86 are generally well-maintained highways with occasional potholes and uneven surfaces, particularly after the monsoon season. Local roads leading into Maniambadi may be narrow and have less consistent maintenance.

## 5. Suggestions for Alternative Routes for Emergencies:

- An alternative route could include taking the NH44, bypassing Salem, though this may increase travel time. Local roads around these major highways can also serve as detours, but their quality and width should be checked in advance.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Transport of hazardous materials requires proper documentation and adherence to Indian Motor Vehicle Rules. There are restrictions on transport through densely populated areas and during certain hours, particularly through urban centers like Salem.

## 7. Overview of Historical Incidents:

- Salem has reported occasional heavy vehicle incidents due to congestion and narrow urban roads. There is also a historical note of chemical spills on state highways; however, these are relatively rare.

## 8. Environmental Considerations and Sensitive Areas:

- Protected forest areas are near Dharmapuri, requiring caution to prevent any spillage or environmental impact. Drivers should maintain lane discipline and avoid overtaking in these stretches.

## 9. Analysis of Communication Coverage:

- Mobile network coverage is generally reliable on major highways such as NH79, but there may be dead zones in rural stretches, particularly as one approaches Odasalahalli.

10. Estimated Emergency Response Times:

- Emergency response within urban areas such as Salem is relatively prompt (15-30 minutes). However, rural segments might experience delays due to road access and lesser frequency of emergency services, potentially extending response times to 45 minutes or more.

11. Overall Summary of Risk Assessment:

- The route has moderate risks due to weather conditions and traffic congestion, particularly in Salem. Proper planning, including checking weather forecasts and avoiding peak traffic hours, will mitigate these risks. Good road infrastructure on national highways supports smooth transit, though local roads require caution. Adequate communication devices are essential, especially when traversing less populated stretches. Familiarity with emergency contacts and clear routes are vital in preparing for incidents involving hazardous materials transport.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 12.13658, 78.16171 | 15 KM/Hr    |
| 1  | Roundabout | High       | 12.14589, 78.17156 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.44900, 77.87406 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.45000, 77.87213 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.45309, 77.87325 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.46765, 77.87037 | 30 KM/Hr    |
| 10 | Turn       | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 11 | Turn       | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 12 | Turn       | High       | 11.49224, 77.89601 | 15 KM/Hr    |
| 13 | Turn       | Medium     | 11.53512, 77.95717 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.53606, 77.95869 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.64183, 78.11810 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 16 | Turn      | Medium     | 11.64338, 78.11998 | 30 KM/Hr    |
| 17 | Turn      | Medium     | 11.91357, 78.05868 | 30 KM/Hr    |
| 18 | Turn      | Medium     | 11.92556, 78.06499 | 30 KM/Hr    |
| 19 | Turn      | Medium     | 11.93694, 78.05425 | 30 KM/Hr    |
| 20 | Turn      | Medium     | 11.95531, 78.04698 | 30 KM/Hr    |
| 21 | Turn      | Medium     | 11.95854, 78.04990 | 30 KM/Hr    |
| 22 | Turn      | Medium     | 11.96015, 78.04945 | 30 KM/Hr    |
| 23 | Turn      | Medium     | 11.96370, 78.05142 | 30 KM/Hr    |
| 24 | Turn      | Medium     | 11.96433, 78.05113 | 30 KM/Hr    |
| 25 | Turn      | High       | 11.96542, 78.04938 | 15 KM/Hr    |
| 26 | Turn      | High       | 11.96650, 78.04927 | 15 KM/Hr    |
| 27 | Turn      | Medium     | 11.96804, 78.05080 | 30 KM/Hr    |
| 28 | Turn      | High       | 12.14045, 78.13962 | 15 KM/Hr    |
| 29 | Turn      | High       | 12.14497, 78.16507 | 15 KM/Hr    |
| 30 | Turn      | Medium     | 12.14557, 78.27689 | 30 KM/Hr    |
| 31 | Turn      | Medium     | 12.14594, 78.27904 | 30 KM/Hr    |

## Emergency Locations

|   | type     | name                    | coordinates               | speed_limit | risk_level |
|---|----------|-------------------------|---------------------------|-------------|------------|
| 0 | hospital | Thilagam Hospital       | 11.4753573,<br>77.8685829 | 30 km/h     | Medium     |
| 1 | clinic   | Meyyazhagan Clinic      | 11.4752254,<br>77.8685671 | 30 km/h     | Medium     |
| 2 | hospital | Dr. Jaganathan Hospital | 11.4752717,<br>77.8709312 | 30 km/h     | Medium     |
| 3 | hospital | Kathir Eye Hospital     | 11.4740976,<br>77.8681591 | 30 km/h     | Medium     |
| 4 | police   | Police Quarters         | 11.4741243,<br>77.8673495 | 30 km/h     | Medium     |
| 5 | hospital | Sankari Hospital        | 11.4777496,<br>77.873892  | 30 km/h     | Medium     |
| 6 | clinic   | Dr. Dhanasekar Clinic   | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 7  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 9  | hospital | C. N. Hospital                                                  | 11.55724,<br>78.0096      | 30 km/h     | Medium     |
| 10 | hospital | Annapoorna Medical College Hospital                             | 11.57693,<br>78.037627    | 30 km/h     | Medium     |
| 11 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 12 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 13 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 16 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 20 | hospital | Bharathi Hospital, Salem                                        | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 21 | hospital | Sisu Hospital Inida Ltd                                         | 11.6505276,<br>78.1205003 | 30 km/h     | Medium     |
| 22 | hospital | Aishwarya Hospital                                              | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 23 | hospital | C.M. Hospital                                                   | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |
| 25 | pharmacy | Healthy Pharmacy                                                | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |
| 26 | pharmacy | Thulasi Pharmacy                                                | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 27 | hospital | Sri Gokulam Speciality Hospital                                 | 11.6788724,<br>78.1274258 | 30 km/h     | Medium     |
| 28 | hospital | Global Medical Center                                           | 11.6867049,<br>78.1186697 | 30 km/h     | Medium     |
| 29 | hospital | Global Medical Centre Jagirammalayam                            | 11.6894409,<br>78.1108847 | 30 km/h     | Medium     |
| 30 | hospital | Manipal Hospital                                                | 11.703072,<br>78.1026612  | 30 km/h     | Medium     |
| 33 | pharmacy | MJ Medicals                                                     | 11.7232728,<br>78.0698965 | 30 km/h     | Medium     |
| 34 | pharmacy | Salem Medicals                                                  | 11.7343192,<br>78.0566844 | 30 km/h     | Medium     |
| 36 | clinic   | Manimuthu Skin Clinic                                           | 11.7521189,<br>78.0484058 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 37 | hospital | St. Joseph's Speciality Hospital   | 11.761611,<br>78.0522344  | 30 km/h     | Medium     |
| 39 | hospital | Anugraha Multi Speciality Hospital | 11.773549,<br>78.0546408  | 30 km/h     | Medium     |
| 41 | hospital | L.R. Hospital                      | 11.8703354,<br>78.0844554 | 30 km/h     | Medium     |
| 42 | clinic   | JSA Clinic                         | 11.8700619,<br>78.0844982 | 30 km/h     | Medium     |
| 43 | hospital | Thoppur,Government Hospital        | 11.9398824,<br>78.0551658 | 30 km/h     | Medium     |
| 45 | clinic   | Bee Tee Clinic                     | 12.138889,<br>78.1576748  | 30 km/h     | Medium     |
| 46 | clinic   | Sheela Clinic                      | 12.138725,<br>78.1573436  | 30 km/h     | Medium     |
| 47 | clinic   | Suguna Clinic                      | 12.1386752,<br>78.1575468 | 30 km/h     | Medium     |
| 48 | hospital | Vijaya Hospital                    | 12.1399758,<br>78.1532115 | 30 km/h     | Medium     |
| 49 | hospital | Navina Hospital                    | 12.1367521,<br>78.1610296 | 30 km/h     | Medium     |
| 50 | clinic   | Bharani Neurologist                | 12.135969,<br>78.16255    | 30 km/h     | Medium     |
| 51 | hospital | Pragadeesh Hospital                | 12.135284,<br>78.1591867  | 30 km/h     | Medium     |
| 52 | hospital | D.N.V Hospital                     | 12.1351328,<br>78.162869  | 30 km/h     | Medium     |
| 53 | clinic   | Ragavendra Physiotherapy Clinic    | 12.135569,<br>78.1635247  | 30 km/h     | Medium     |
| 54 | hospital | Kamatchi Hospital                  | 12.1350608,<br>78.1594841 | 30 km/h     | Medium     |
| 55 | clinic   | Malar Siddha Clinic                | 12.1350485,<br>78.159226  | 30 km/h     | Medium     |
| 56 | hospital | Kamala Nehru Hospital              | 12.13404,<br>78.1633476   | 30 km/h     | Medium     |
| 57 | clinic   | Govindaraj Clinic                  | 12.1344914,<br>78.1642372 | 30 km/h     | Medium     |
| 58 | hospital | Sri Priya Hospital                 | 12.135347,<br>78.164391   | 30 km/h     | Medium     |



Crowded Spots

|    | type       | name                                                    | coordinates            | speed_limit | risk_level |
|----|------------|---------------------------------------------------------|------------------------|-------------|------------|
| 14 | university | Vinayaka Missions Vinayaka Missions Research Foundation | 11.5974056, 78.0843617 | 30 km/h     | Medium     |
| 15 | school     | Vinayaka Missions Annapoorani Matric School             | 11.5971662, 78.0839876 | 30 km/h     | Medium     |
| 17 | school     | Dist Education Training                                 | 11.6133799, 78.1022409 | 30 km/h     | Medium     |
| 18 | school     | DIET - Salem                                            | 11.6144158, 78.1025078 | 30 km/h     | Medium     |
| 19 | school     | Diamond Rays Matric HR Sec School                       | 11.6385629, 78.1206259 | 30 km/h     | Medium     |
| 24 | college    | Sona College                                            | 11.6768, 78.1257052    | 30 km/h     | Medium     |
| 31 | college    | Govt Engineering College                                | 11.7128537, 78.0877084 | 30 km/h     | Medium     |
| 32 | university | Periyar University                                      | 11.7186758, 78.0771089 | 30 km/h     | Medium     |
| 35 | school     | Fathima Hr Sec School                                   | 11.7410508, 78.0479272 | 30 km/h     | Medium     |
| 38 | school     | John Britto Matriculation Higher Secondary School       | 11.7676907, 78.0513092 | 30 km/h     | Medium     |
| 40 | college    | Dhirajlal Gandhi College of Technology                  | 11.7968014, 78.0562095 | 30 km/h     | Medium     |
| 44 | college    | Laxminarayana Polytechnic College                       | 12.0421277, 78.103203  | 30 km/h     | Medium     |

Route Photos of Risky Spots